routers, and firewalls that route the network traffic or restrict them. Various network operations such as control, monitoring, and management have dedicated hardware devices connected to serve the traditional network. But this traditional network has fallen short due to the growing network size and demands of computing (such as data centres) as well as the network needs of large scale software applications. Setting up switches and routers has become too complicated and error-prone. Implementing and configuring network policies on such a cluttered network with so many static devices has become burdensome and complex.

Traditional network devices have always been criticised for vertical integration. Because of this, the network complexity, management, and operation cost increase whenever new services, technologies, or hardware are adapted to be deployed in the network. In vertical integration, the control plane and the data plane are packed inside these network devices.

The control plane is designated to fill the routing table based on the topology discovery. The routing table decides where to send the incoming data packets. The function of the control plane is to provide the routing and signal protocols to be used. The data plane or forwarding plane, on the other hand, forwards the data packets once the control plane has decided what to do with the given packet. In Figure 1, we can see how the control plane and data plane work while being vertically integrated into the traditional networking device switch.

To cater to all these shortcomings of traditional networking, software defined networking (SDN) intelligently decouples the control plane and the data plane. SDN promises a new networking environment that knows no limits and bounds, where network management is easy and enables network configuration programmatically. With SDN, we are smartly converting our static architecture of traditional networks to a more dynamic architecture. When we disassociate the control plane and data plane, we are making an intelligent unit of networking, i.e., control plane

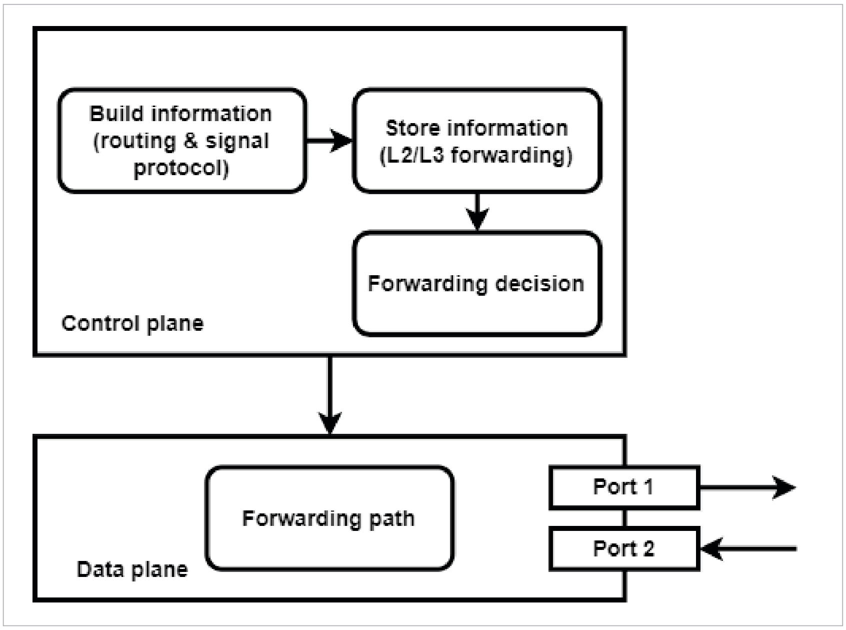

Figure 1: Control and data planes in a traditional network

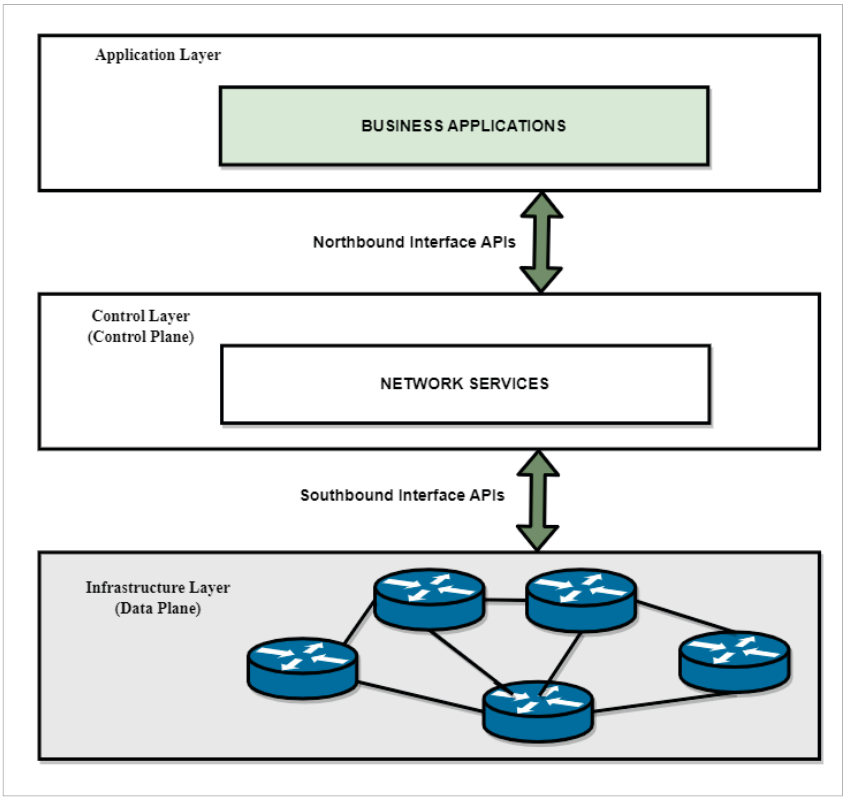

Figure 2: SDN architecture